

SHANGCHUAN DAO, China

WMO: 596730

Lat: 21.7358N

Lon: 112.7731E

Elev: 18

StdP: 101.11

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.0	9.3	-0.9	3.5	12.7	1.4	4.2	12.8	17.8	9.2	15.8	10.4	11.5	20	0.430

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.7	32.6	27.3	31.9	27.1	31.3	26.9	28.2	30.8	27.9	30.4	27.6	30.1	3.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.6	23.6	29.7	27.2	23.0	29.5	27.0	22.7	29.4	91.2	30.9	89.7	30.5	88.4	30.2	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.0	12.7	10.7	DB	6.9	34.4	1.7	0.8	5.7	35.0	4.7	35.4	3.8	35.8	2.5	36.4	(4)
			WB	4.0	28.9	1.6	0.6	2.9	29.3	2.0	29.7	1.1	30.0	-0.1	30.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.4	15.7	16.8	19.6	23.4	26.6	28.4	28.8	28.6	27.8	25.6	21.8	17.5
	DBStd	5.37	3.36	4.02	3.20	2.79	2.04	1.46	1.29	1.41	1.57	1.84	2.80	2.97
	HDD10.0	6	2	3	0	0	0	0	0	0	0	0	0	1
	HDD18.3	243	92	68	24	3	0	0	0	0	0	0	7	50
	CDD10.0	4909	180	194	298	401	516	553	584	578	534	483	353	234
	CDD18.3	2105	11	27	64	153	258	303	326	319	284	225	110	25
	CDH23.3	19824	2	22	119	813	2418	3554	3958	3798	3081	1691	349	19
CDH26.7	6174	0	0	2	93	619	1301	1543	1411	943	245	17	0	
Wind	WSAvg	4.7	5.9	5.3	4.6	4.1	3.9	3.9	3.8	3.5	4.4	5.2	5.8	6.4
Precipitation	PrecAvg	2159	27	47	74	175	347	341	269	345	315	185	40	28
	PrecMax	2952	89	265	380	448	971	842	605	877	904	517	200	181
	PrecMin	1028	0	0	5	3	34	45	91	100	28	0	0	0
	PrecStd	506	28	63	76	126	231	193	143	189	198	144	49	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.6	25.2	26.8	29.7	32.1	32.9	33.5	33.7	32.9	30.8	28.3	25.2
		MCWB	20.0	22.0	23.6	25.6	26.9	27.2	28.1	27.6	27.2	25.4	23.9	21.5
	2%	DB	22.2	24.0	25.7	28.7	30.9	31.7	32.3	32.6	31.8	29.7	27.1	23.7
		MCWB	19.0	21.6	23.2	25.5	26.7	27.1	27.4	27.4	26.6	25.0	23.0	20.7
	5%	DB	21.1	22.9	24.7	27.8	30.1	31.2	31.6	31.8	31.0	28.8	26.1	22.6
		MCWB	18.2	20.9	22.7	25.0	26.5	27.0	27.1	27.1	26.4	24.3	22.6	19.8
	10%	DB	20.0	21.7	23.7	27.0	29.4	30.6	31.0	31.0	30.2	28.0	25.3	21.5
		MCWB	17.6	19.6	21.9	24.8	26.3	27.0	27.0	27.0	26.0	23.8	22.1	18.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.1	23.2	24.4	26.7	28.0	28.5	28.7	28.6	28.2	26.7	25.0	23.2
		MCDB	22.6	24.2	25.9	28.2	29.8	30.6	32.1	31.8	31.1	28.9	26.8	24.1
	2%	WB	19.8	22.3	23.7	26.1	27.6	28.1	28.1	28.1	27.6	26.0	24.3	21.8
		MCDB	21.4	23.6	25.1	27.8	29.5	30.2	30.9	31.0	30.3	28.4	26.1	22.9
	5%	WB	18.9	21.1	23.1	25.6	27.3	27.7	27.7	27.7	27.2	25.5	23.7	20.4
		MCDB	20.6	22.6	24.4	27.3	29.3	29.9	30.3	30.5	29.7	27.8	25.6	22.1
	10%	WB	18.0	20.0	22.3	25.0	26.8	27.5	27.5	27.4	26.8	25.0	23.0	19.0
		MCDB	19.6	21.3	23.4	26.7	28.9	29.7	30.0	30.0	29.2	27.3	25.0	21.1
Mean Daily Temperature Range	5% DB	MDBR	4.3	4.0	3.9	3.6	3.7	3.5	3.7	4.0	4.0	3.8	4.2	4.3
		MCDBR	5.3	5.2	4.7	4.5	4.4	4.2	4.6	5.3	5.0	4.5	4.7	4.9
		MCWBR	2.8	2.8	2.6	2.0	1.7	1.5	1.7	2.1	2.1	2.1	2.3	2.7
	5% WB	MCDBR	4.9	4.5	4.4	4.2	3.8	3.5	4.0	4.6	4.5	4.0	3.9	4.3
		MCWBR	3.1	3.0	2.9	2.2	1.7	1.7	1.8	2.2	2.2	2.0	2.3	3.0
Clear-Sky Solar Irradiance	taub	0.592	0.631	0.761	0.730	0.555	0.522	0.508	0.579	0.625	0.696	0.593	0.590	
	taud	1.784	1.732	1.523	1.620	2.057	2.173	2.205	1.976	1.838	1.643	1.834	1.805	
	Ebn at Noon	673	680	615	647	766	785	797	744	702	621	667	655	
	Edh at Noon	203	226	289	266	171	151	146	184	208	244	191	193	
All-Sky Solar Radiation	RadAvg	2.96	3.07	3.33	3.86	4.66	4.80	5.38	4.96	4.66	4.30	3.55	3.11	
	RadStd	0.49	0.48	0.43	0.59	0.48	0.70	0.60	0.40	0.41	0.36	0.39	0.43	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+0.26	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	N/A	N/A	N/A	+0.20	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page